INTER-PROCESS COMMUNICATION USING PIPE

Muthukutti R **241801175**

Write a C program to demonstrate Inter-Process Communication (IPC) using a pipe. The program should create a parent and child process using `fork()`. The parent process should send a message to the child process through the pipe, and the child process should read and display the received message.

Input Format:

The input consists of a single integer `n` entered by the user.

Output Format:

- Displays the value sent by the parent process
- Displays the processed value (double of the number) received and computed by the child process

Sample Input:

5

Sample Output:

Parent sent: 5
Child received: 5
Child processed value (double): 10

CODE:

```
#include <stdio.h>

#include <unistd.h>

#include <string.h>
```

```
int main() {
    int fd[2];
    pid_t pid;

    char write_msg[] = "Hello from Parent Process";
    char read_msg[100];

    // Create pipe
    if (pipe(fd) == -1) {
        printf("Pipe failed\n");
        return 1;
    }

    // Create child process
    pid = fork();

    if (pid < 0) {
        printf("Fork failed\n");
        return 1;
    }
```

```
    // Parent Process
    if (pid > 0) {
        close(fd[0]); // Close read end

        write(fd[1], write_msg, strlen(write_msg) + 1);
        printf("Parent sent: %s\n", write_msg);

        close(fd[1]); // Close write end
    }

    // Child Process
    else {
        close(fd[1]); // Close write end

        read(fd[0], read_msg, sizeof(read_msg));
        printf("Child received: %s\n", read_msg);

        close(fd[0]); // Close read end
    }

    return 0;
}
```

Output:

Enter a number: 5
Parent sent: 5
Child received: 5
Child processed value (double): 10